Figure 9.9 Joby eVTOL cutaway diagram (propulsion and safety systems). ⏎

9.8.10 United States: Quantum-resilient UAV communication architecture

The United States has been actively developing quantum-resilient communication architectures for unmanned aerial systems to protect them against emerging cyber and electronic warfare threats. As drones become increasingly autonomous and network-dependent, traditional encryption and radiobased control links face rising vulnerabilities – especially with the anticipated future capabilities of quantum computing, which could rapidly break classical cryptographic algorithms. To mitigate these risks, US defense research institutions such as the Air Force Research Laboratory (AFRL), DARPA, and MIT Lincoln Laboratory have introduced secure communication frameworks designed to ensure confidentiality, integrity, and continuity of UAV operations even in contested electromagnetic environments.

A core strategy of this architecture is the integration of post-quan tum cryptographic algorithms, which rely on mathematically complex lattice-based encryption that remains resistant to quantum decryption attacks. In addition, experimental systems are incorporating quantum key distribution (QKD), enabling drones and ground control stations to exchange encryption keys using the quantum properties of photons. Any attempt to intercept or tamper with these keys results in detectable disturbances, allowing the system to identify intrusions in real time and switch to alternate secure channels.

The US approach also emphasizes decentralized mesh networking among UAV fleets, enabling drones to maintain operational coordination even if satellite links or command hubs are jammed. In this configuration, individual UAVs function as autonomous communication nodes, relaying